SpecVision AI

AI-Powered Product Specification and Cost Estimation for Custom Jewelry

What this is

An ML engineering system for the custom jewelry domain. It takes a short sales brief (and optionally a reference photo of a ring) and produces: a structured product specification, a ranked match against a 42-item inventory catalog, a classical computer-vision estimate of visible gemstones in the photo, an indicative price, and an explicit confidence/review summary. The core design goal is honesty about uncertainty: every field carries a source (stated by the user / inferred by the model / unknown) and a confidence level, and the system is built to say "unknown" or "needs review" rather than present a guess as a confident fact.

What was built

- Backend (FastAPI, Python): LLM-based extraction (Groq, live mode) with a fully functional deterministic offline fallback -- no API key required to run the whole pipeline.
- Deterministic inventory matching: a weighted attribute-scoring engine (no LLM) that ranks the catalog, with a documented 0.72 auto-selection threshold below which it explicitly declines to claim a reliable match.
- Classical computer vision (OpenCV, no paid vision API): Hough-circle + contour hybrid pipeline that estimates visible gemstone count, with its own confidence assessment and image-quality warnings.
- Pricing engine: indicative price derived from the matched catalog item plus documented adjustments, or explicitly "unavailable" when there is nothing reliable to price from.
- Confidence & review layer: cross-checks the CV estimate against the brief, runs deterministic plausibility checks (implausible carat, metal/purity contradictions), and raises stable, actionable review flags instead of hiding uncertainty.
- React/TypeScript frontend consuming the real API contract (no mocked data), with distinct live-mode and offline-mode indicators.
- A 16-case labeled evaluation harness using real photographs (not synthetic), a 52-test automated pytest suite, and a documented failure analysis of the three worst observed cases with root causes and proposed fixes (EVALUATION.md).

How to run it

cd backend && .venv\Scripts\Activate.ps1 && uvicorn app.main:app --reload --port 8000
cd frontend && npm run dev (opens at http://localhost:5173)

Full setup instructions, environment variables, and architecture are in README.md. Measured evaluation metrics and failure analysis are in EVALUATION.md.

Test case 1 -- successful end-to-end match

Brief: "A solitaire ring in 18k yellow gold with a round diamond center stone, about 0.45 carats, tension setting, eight moissanite side stones, band width 1.7mm." (no photo)

Mode: offline/demo (deterministic, no API call -- fully reproducible). Run with:

python evaluation/runner/run_evaluation.py --mode offline

Extracted specification

Product type	ring	**HIGH**
Design category	solitaire	**HIGH**
Metal	18k yellow gold	**HIGH**
Setting type	tension	**HIGH**
Center stone	round diamond, 0.45ct	**HIGH**
Side stones	8x moissanite	**HIGH**
Band width	1.7mm	**HIGH**

Inventory match

Selected: SV-001 -- "Solitaire Round Yellow Gold Ring" -- match score 100% (threshold: 72%) -- PASSED, auto-selected.

Indicative price

USD 2,568.00 (base catalog price, no adjustments needed -- spec was within the matched item's carat/side-stone range). Formula: indicative_total = catalog_unit_price + extra_carat_adjustment(if spec carat exceeds catalog max) + extra_side_stone_adjustment(if spec side count exceeds catalog count)

Confidence & review

Status: AUTO APPROVED

No review flags were raised -- every stated attribute was extracted correctly and matched a real catalog item within tolerance.

Test case 2 -- honest reporting of a real CV limitation

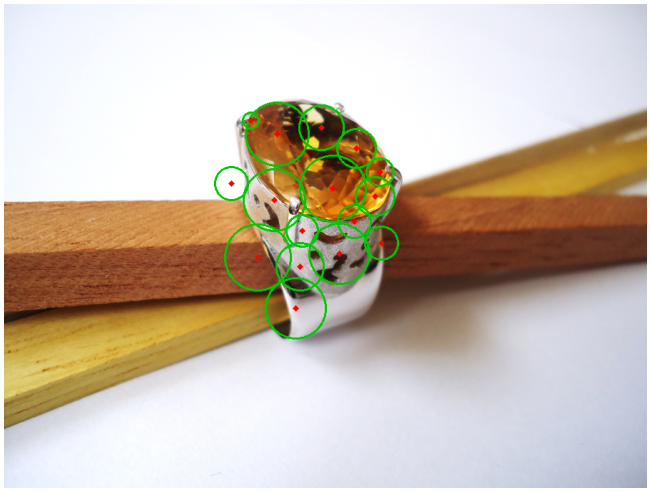
Brief: "A sterling silver ring with a large oval citrine centerpiece and an open filigree band, no side stones." (real photo, Wikimedia Commons CC BY-SA, see evaluation/images/PROVENANCE.md)

Mode: offline/demo. This exact case is analyzed in depth as "Failure #1" in EVALUATION.md.

Original photo



CV detections overlaid (green circles)



Extracted specification

Metal	925 sterling silver	HIGH
Center stone type	Unknown	LOW
Center stone shape	oval	HIGH
Side stones	0	HIGH

"Citrine" is not in the system's controlled gemstone vocabulary, so center_stone_type is correctly left Unknown rather than guessed -- a deliberate, documented scope limitation, not a crash.

Computer vision result

Visible components detected	16	HIGH

The photo's pierced filigree metalwork was mistaken for 16 small gemstones by the classical detector, with HIGH self-reported confidence and no image-quality warning -- the ring actually has zero side stones. This is a real, known limitation of the classical CV baseline (filigree/pierced metalwork resembles a ring of small stones), documented in EVALUATION.md as the project's #1 worst observed failure, with a proposed fix (geometric-regularity check).

Why the system still catches the error

Because the brief explicitly states "no side stones," the confidence layer cross-checks the CV count against the brief and finds a disagreement ratio far above the 40% threshold -- so this case is correctly flagged for human review despite the CV module's own overconfidence. If the same photo were paired with a brief that simply didn't mention side-stone count, this cross-check would not fire -- a limitation stated explicitly in EVALUATION.md, not hidden.

Confidence & review

Status: NEEDS REVIEW

[WARNING] CV_BRIEF_COUNT_DISAGREEMENT

CV detected 16 visible component(s) vs. 0 implied by the brief (relative difference 1600%).

[WARNING] INCOMPLETE_SPEC_FOR_PRICING

Specification is missing or uncertain in fields that affect pricing: center_stone.gemstone_type, center_stone.carat.

[WARNING] NO_RELIABLE_INVENTORY_MATCH

No candidate reached the auto-selection threshold of 0.72 (best score: 0.61). A high similarity score does not prove exact product

equivalence; results below threshold are shown for reference only and require human review.

[WARNING] PRICE_UNAVAILABLE

No inventory candidate met the auto-selection matching threshold, so there is no defensible catalog reference to price from. An indicative price would be a guess.

Measured evaluation results (16 labeled cases)

Full methodology, per-field breakdowns, and the three-worst-failure analysis are in EVALUATION.md. Figures below are read directly from evaluation/results/evaluation_offline.json -- reproducible with: python evaluation/runner/run_evaluation.py --mode offline

Extraction accuracy (offline / deterministic mode)

- Categorical fields (metal type, purity, color, setting, stone type/shape): 92-100% exact-match accuracy on cases with a known gold value; design_category 71% (documented taxonomy-overlap limitation).
- 0 fabricated values across every field where the correct answer was "unknown."

Inventory matching

- Top-1 / Top-3 hit rate: 100% on the one case engineered to match a specific catalog item.
- No-reliable-match handling: 100% correct (8/8 cases engineered to match nothing reliably).

Computer vision

- Mean absolute error: 16.71 stones across 7 real photos -- driven almost entirely by two explainable, documented failure modes (filigree metalwork and faceted-stone reflections), not a uniform small error. See EVALUATION.md for the per-image breakdown.

Review behavior

- Precision: 93% | Recall: 100% | Flag rate: 88% (14/16 cases correctly got at least one actionable flag).

Operational

- 52/52 automated tests passing. 16/16 evaluation cases executed successfully, 0 crashes.
- Live-mode run: 15 real Groq API calls, $0.0000 cost (free tier), including one real HTTP 429 rate-limit recovery and one real malformed-JSON repair -- both handled correctly.